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2. MATLAB matrices and their elements

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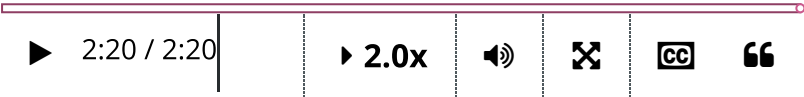
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 Calculator

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Accessing elements of a matrix



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...element vector of our column indices, which we enclose in parentheses.

And look at that!

The result is a two by three matrix containing exactly the subset of data we wanted.

When analyzing data, you might find yourself needing to extract an entire row or column.

Extracting a whole row or column of data is so common, there's a shortcut.

A single colon references all of the elements in that dimension together.

That's better.

Here, the colon in the row dimension indicates that we want all of the rows

Entries of a matrix (External resource) (1.0 points possible)

Entries of a matrix

1. Suppose we have created a matrix **A**. Then we can extract the element of the matrix by defining it as a variable *a*:

```
a = A(m,n)
```

For example, if **A** is the following matrix

$$\mathbf{A} = \begin{bmatrix} -1 & 1 \\ 3 & 2 \end{bmatrix}$$

then the command

```
a = A(2,1)
```

will define a variable *a* with the value 3.

2. We can also extract the columns (or rows) of a matrix. The command

```
v = A(:,1);
```

will create a column vector whose elements are the same as those in the first column of **A**.

$$\mathbf{v} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

Now do the following. Use the script below to create a 3×3 matrix **B** whose entries are

```
B = randi([0,50],3);
```

Then:

1. Define a new variable *b* as the element in the 2nd row and 3rd column of **B**.
2. Define a new row vector *v* which corresponds to the second row of **B**.

2. Define a new row vector v , which corresponds to the second row of B .

You should use the same syntax explained above. You will not get full credit for simply t

Script ?

Save Reset

MATLAB Documentation (https://www.mathworks.com/help/)

```
1 % Firstly enter a 3x3 matrix B of random integers
2 B = randi([0,50],3);
3 % Now define b and v
4 b = B(2,3);
5 v = B(2,:);
```

Run Script ?

Assessment: All Tests Passed

Submit ?

Value of b

Value of v

Output

Code ran without output.

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